

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Civil Engineering
MID SEMESTER EXAMINATION, March 2023

B.Tech.Semester-4
Course Name: Highways and Airport Engineering
Time allowed: 2 Hours

Branch: Civil Engineering
Course Code: CE43106
Full Marks:30

- Instructions:
- (i) Answer all questions.
 - (ii) All questions carry the same marks.
 - (iii) Answer each question in your own words.
 - (iv) All parts of the answer must be in the same place.
 - (v) Use of graph paper is allowed.

		Marks	CO	BL
Q1.(i)	What are the advantages and disadvantages of road transportation over other mode of transportation? <i>Accidents</i> <i>Organisation</i>	3	1	II
(ii)	Sketch the normal cross-section of a highway in rural areas in embankment showing the different elements with approximate dimensions.	3	2	V
Q2.(i)	What are the basic properties which the bitumen should possess for road construction?	3	1	II
(ii)	The modulus of subgrade reaction of soil tested with a 350mm diameter rigid plate is 5.25MPa/mm. Estimate the modulus of subgrade reaction corresponding to standard plate diameter. Also, estimate the pressure that would cause a plate settlement of 1.25mm when the 750mm diameter plate is used and Elastic Modulus of subgrade soil.	3		
			$\gamma = \frac{E_s}{1.18 \times a}$	
Q3.(i)	How the vehicle dimensions affect the geometric elements of the Highways?	2	2	V
(ii)	Design a template for parabolic camber for a two-lane bituminous concrete pavement in a heavy rainfall area. Draw a neat sketch.	4		
Q4.	Two cars are approaching from the opposite direction on gradient of single lane two-way State Highways.	6	2	V
	Car A	Car B		
Travelling speed =	60kmph	80kmph		
Gradient =	+ 2%	-2%		
Brake efficiency =	80%	90%		
Reaction time =	1.5secs	2.5secs		
Coefficient of longitudinal friction =	0.34			
			$y = \frac{2x^2}{w\pi}$	
			0.64	

Calculate (i) Stopping sight distance (SSD)

(ii) Head-on collision distance

(iii) Head light sight distance

Q5.(i) What are the requirements for a good coarse aggregate for road construction? State their suitability? 2 1 II

(ii) Determine CBR and Resilient Modulus of the subgrade soil using the CBR test readings: 4

10.6 x 10⁶

Penetration (mm)	Load (kg)
0	0
0.5	2
1.0	10
1.5	25
2.0	40
2.5	50
3.0	58
4.0	70
5.0	77.5
7.5	93.2
10.0	102.5
12.0	110.8